

GreenNet III: Source-to-Solution Supervision

Supervise the Green operator action, not pointwise exact-kernel labels

No pointwise Green-kernel labels.

$$w(t) \sim \mathcal{GP}(0, k_\ell)$$

→

$$v(t) = w(t) - ((1-t)w(0) + tw(1))$$

SO $v(0) = v(1) = 0$

→

$$f_{\text{unit}} = \mathcal{L}_{\text{unit}} v$$

$$v_\theta(t) = \int_0^1 G_\theta(t, \eta) f_{\text{unit}}(\eta) d\eta.$$

$$\mathcal{J}_{\text{Green}} \sim \mathbb{E} \left[\int_0^1 |v_\theta(t) - v(t)|^2 dt \right].$$

Training checks whether the learned Green operator maps sources generated from target solutions back to those target solutions.